

Example 1:

$$\max z = 6x_1 + 5x_2$$

$$\text{s.t. } x_1 + x_2 \leq 5$$

$$3x_1 + 2x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

$$\max z = 6x_1 + 5x_2$$

$$\text{s.t. } x_1 + x_2 + s_1 = 5$$

$$3x_1 + 2x_2 + s_2 = 12$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Simplex Requirements.

$$1) \text{ } +RHS \Rightarrow \checkmark$$

$$2) \text{ } = \checkmark$$

$$3) \text{ } \geq 0 \checkmark$$



$$\max z = 6x_1 + 5x_2$$

$$\text{s.t. } x_1 + x_2 + s_1 = 5$$

$$3x_1 + 2x_2 + s_2 = 12$$

$$x_1, x_2, s_1, s_2 \geq 0$$

starting b.f.s

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(s_1, s_2) = (5, 12)$$

basic.

$$x_1 = x_2 = 0$$

nonbasic

Basic	x_1	x_2	s_1	s_2	R.H.S
z	-6	-5	0	0	0
s_1	1	1	1	0	5
s_2	3	2	0	1	12
z	0	-1	0	2	24
s_1	0	1/3	1	-1/3	1
x_1	1	2/3	0	1/3	4

max $\rightarrow (-) z$ -row.

$$x_1 = \text{E.V.}$$

min Ratio

$$5/1 = 5$$

$$12/3 = 4$$

$$\Rightarrow s_2 = \text{L.V.}$$

$$(-1) \left(\text{New } x_1\text{-row} \right) + s_1\text{-row} = \text{New } s_1\text{-row.}$$

$$\begin{array}{ccccc} -1 & -\frac{2}{3} & 0 & -\frac{1}{3} & -4 \end{array}$$

$$\begin{array}{ccccc} + & 1 & 1 & 1 & 0 & 5 \\ \hline 0 & \frac{1}{3} & 1 & -\frac{1}{3} & 1 \end{array}$$

$$6 \left(\text{New } x_1\text{-row} \right) + \left(\text{old } z\text{-row} \right) = \text{New } z\text{-row}$$

$$\begin{array}{ccccc} 6 & 4 & 0 & 2 & 24 \end{array}$$

$$\begin{array}{ccccc} -6 & -5 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{ccccc} + & & & & \\ \hline 0 & -1 & 0 & 2 & 24 \end{array}$$

	x_1	x_2	s_1	s_2	R.H.S
z	0	-1	0	2	24
s_1	0	1/3	1	-1/3	1
x_1	1	2/3	0	1/3	4

$x_2 = \text{E.V.}$

$$1/1/3 = 3 \Rightarrow s_1 = \text{L.V.}$$

$$4/2/1 = 6$$

(x3)

s_1	0	$1/3$	1	$-1/3$	4	Ratio
x_1	1	$2/3$	0	$1/3$	4	$1/(1/3) = 3 \Rightarrow s_1: LV.$
z	0	0	3	1	27	$4/(2/3) = 6$
x_2	0	1	3	-1	3	
x_1	1	0	-2	1	2	

$$\frac{-2}{3} \left(\text{New } x_2\text{-row} \right) + x_1\text{-row} = \text{New } x_1\text{-row}.$$

$$\text{New } x_2\text{-row} + 0 \downarrow z\text{-row} = \text{New } z\text{-row}$$

$$\begin{array}{rrrrr} 0 & -\frac{2}{3} & -2 & \frac{2}{3} & -2 \\ + & 1 & \frac{2}{3} & 0 & \frac{1}{3} & 4 \\ \hline 1 & 0 & -2 & 1 & 2 \end{array}$$

$$\begin{array}{rrrrr} 0 & 1 & 3 & -1 & 3 \\ + & 0 & -1 & 0 & 2 & 24 \\ \hline \end{array}$$

No entering variable.

Since all z -row coefficients are non-negative, we reached the optimum solution.

$$x_1^* = 2$$

$$s_1^* = 0$$

$$z^* = 27.$$

$$x_2^* = 3$$

$$s_2^* = 0$$

Example 2:

$$\min z = 2x_1 + 7x_2 - 2x_3$$

$$\text{s.t. } x_1 + x_2 + x_3 \leq 1$$

$$-4x_1 - 2x_2 + 3x_3 \leq 2$$

$$x_1, x_2, x_3 \geq 0$$

$$\begin{aligned} \Rightarrow \min z &= 2x_1 + 7x_2 - 2x_3 \\ \text{s.t. } x_1 + x_2 + x_3 + s_1 &= 1 \\ -4x_1 - 2x_2 + 3x_3 + s_2 &= 2 \\ x_1, x_2, x_3, s_1, s_2 &\geq 0. \end{aligned}$$

Starting b.f.s = $(s_1, s_2) = (1, 2)$

nonbasic = $x_1, x_2, x_3 = 0$.

Basic	x_1	x_2	x_3	S_1	S_2	Sol	
z	-2	-7	$2 \Rightarrow 0$	0	0	0	$\min(+)$ in z -row. $x_3 = \text{E.V.}$
S_1	1	1	$1 \Rightarrow 0$	1	0	1	$\min \text{ ratio}$ $1/1 = 1$
S_2	-4	-2	3	0	1	2	$2/3 = 2/3 \Rightarrow$
z	$2/3$	$-17/3$	0	0	$-2/3$	$-4/3$	
S_1	$7/3$	$5/3$	0	1	$-1/3$	$1/3$	
x_3	$-4/3$	$-2/3$	1	0	$1/3$	$2/3$	

$$\begin{array}{cccccc}
 +8 & 4 & -2 & 0 & -2 & -4 \\
 3 & 3 & & & 3 & 3 \\
 + & -2 & -7 & 2 & 0 & 0 & 0 \\
 \hline
 2 & -17 & 0 & 0 & -2 & -4 \\
 3 & 3 & & & 3 & 3
 \end{array}$$

z	$2/3$	$-17/3$	0	0	$-2/3$	$-4/3$	
S_1	$7/3$	$5/3$	0	1	$-1/3$	$1/3$	$\min \text{ ratio}$ $1/3 / 5/3 = 1/5 \Rightarrow$ $S_1 = \text{L.V.}$
x_3	$-4/3$	$-2/3$	1	0	$1/3$	$2/3$	
z	0	$-43/7$	0	$-2/7$	$-4/7$	$-10/7$	
x_1	1	$5/7$	0	$3/7$	$-1/7$	$1/7$	
x_3	0	$2/7$	1	$4/7$	$1/7$	$6/7$	

(New x_1 -row) $\cdot \begin{pmatrix} -2 \\ 3 \end{pmatrix} + \text{old } z\text{-row} = \text{New } z\text{-row}.$

$$\begin{array}{cccccc}
 -2 & -10 & 0 & -2 & 2 & -2 \\
 3 & 21 & & 7 & 21 & 21
 \end{array}$$

$$\begin{array}{cccccc}
 + & \frac{2}{3} & \frac{-17}{3} & 0 & 0 & \frac{-2}{3} & \frac{-4}{3} \\
 \hline
 0 & \frac{-129-43}{21} = \frac{-172}{21} & 0 & \frac{-2}{7} & \frac{-12}{21} & \frac{-30}{21}
 \end{array}$$

$\left(\begin{array}{l} \text{New} \\ x_1\text{-row} \end{array} \right) \cdot \frac{4}{3} + x_3\text{-row} = \text{New } x_3\text{-row.}$

$$\begin{array}{cccccc}
 \frac{4}{3} & \frac{20}{21} & 0 & \frac{4}{7} & \frac{-4}{21} & \frac{4}{21} \\
 + & \frac{-4}{3} & \frac{-2}{3} & 1 & 0 & \frac{1}{3} & \frac{2}{3} \\
 \hline
 0 & \frac{6}{21} & 1 & \frac{4}{7} & \frac{3}{21} & \frac{14}{21}
 \end{array}$$

Since all z-row values are nonpositive, we reached the optimum solution.

$$x_1^* = \frac{1}{7} \quad x_3^* = \frac{6}{7} \quad x_2^* = 0$$

$$S_1^* = S_2^* = 0 \quad z^* = -\frac{10}{7}$$

Simplex Algorithm.

$\swarrow$
max

z-row values (-)

$\searrow$
min.

(+) (E.V)

min ratio rule is
the same

RHS / coefficient
in pivot column.

$\Rightarrow$ min. L.V.

$\swarrow$
(-), 0.

$\times$ do not
calculate
the ratio.